

DEPARTMENT OF CSE

COURSE CODE : CSE 402

COURSE TITLE : Operating System Lab

REPORT : 01

REPORT NAME : FCFS CPU Scheduling

SUBMITTED BY:

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SECTION : C-1

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Problem Statement

Calculate Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), Average Turnaround Time and Average Waiting Time using the FCFS scheduling algorithm. Also represent the process execution sequence.

Given Data

Process	AT	BT
P1	3	5
P2	2	4
P3	4	3
P4	1	2
P5	5	3

Objective

CPU scheduling algorithm and calculate CT, TAT and WT for the given processes. The average turnaround time, average waiting time and the execution sequence are also determined. The objective of this lab is to understand the First Come First Serve (FCFS)

Working Principle

In FCFS scheduling, the process that arrives first gets the CPU first. The processes are arranged according to their arrival time. Completion Time is found from the execution order. Then TAT is calculated by CT - AT and WT is calculated by TAT - BT.

Source Code

```
processes = [
    ["P1", 3, 5],
    ["P2", 2, 4],
    ["P3", 4, 3],
    ["P4", 1, 2],
    ["P5", 5, 3]
]
order = sorted(processes, key=lambda x: x[1])
result = {}
current_time = 0
execution = []

for p in order:
pid = p[0]
at = p[1]
bt = p[2]
 if current_time < at:
current_time = at
ct = current_time + bt
tat = ct - at
wt = tat -bt
current_time = ct
result[pid] = [ct, tat, wt]
execution.append(pid)


print("P_ID AT BT CT TAT WT")
for p in processes:
```

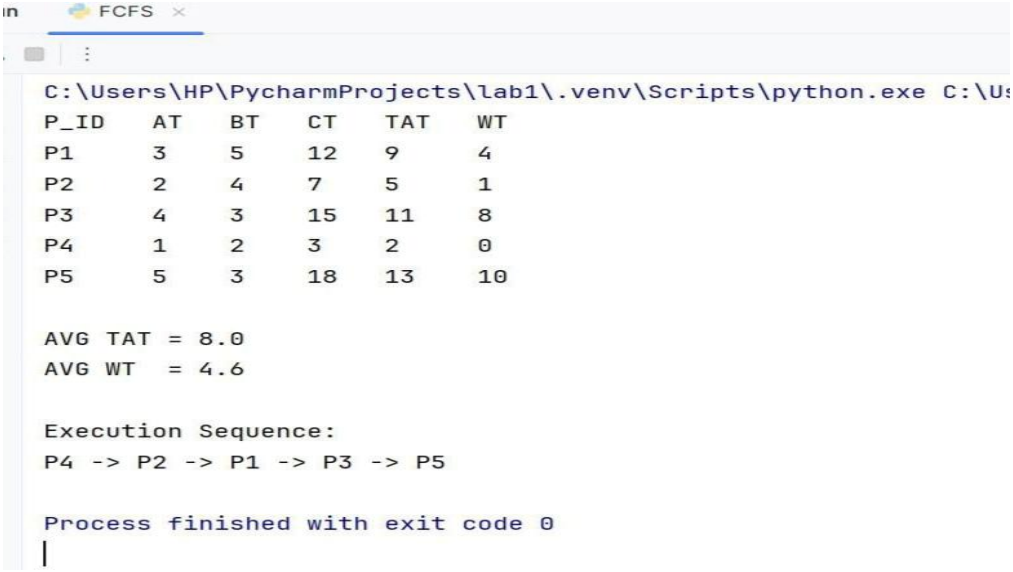
```
pid = p[0]
at = p[1]
bt = p[2]
ct = result[pid][0]
tat = result[pid][1]
wt = result[pid][2]

print(f"{pid:<7}{at:<5}{bt:<5}{ct:<5}{tat:<6}{wt}")
total_tat = 0
total_wt = 0

for p in processes:
total_tat += result[p[0]][1]
total_wt += result[p[0]][2]
avg_tat = total_tat / len(processes)
avg_wt = total_wt / len(processes)

print()
print("AVG TAT =", avg_tat)
print("AVG WT =", avg_wt)
print()
print("Execution Sequence:")
print(" -> ".join(execution))
```

Output



Discussion

In FCFS scheduling, processes are handled based on the order in which they arrive. For the provided process data, the execution sequence was P4, P2, P1, P3, and P5. The Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) were determined for every process. The calculated average TAT was 8.0, while the average WT was 4.6. Through this experiment, the behavior of the FCFS scheduling method was understood, particularly how the arrival time of a process influences its execution sequence and waiting period.

Conclusion

The FCFS CPU scheduling method was successfully implemented using Python. All necessary values, including CT, TAT, WT, average TAT, average WT, and the process execution order, were calculated successfully. This experiment provided a clear understanding of the basic concept of FCFS scheduling. It was also observed that a process is selected according to its arrival order and must remain in the waiting state until the preceding process completes its execution.

GitHub Link